Chapter 17

The Beginning of the Life Cycle

1. Prenatal Development and Care
2. Heredity and Genetics
3. Birth Through Childhood

Lesson 1 — Prenatal Development and Care

Fertilization (conception) occurs when a sperm and egg unite, typically in the fallopian tube, creating a **zygote** with 46 chromosomes (23 from each parent). Within 6 days, **implantation** occurs as the cluster burrows into the uterine wall. Development stages: zygote → embryo (weeks 3–8) → fetus (week 9 through birth).

Three Trimesters of Development

1st Trimester (0–14 wks)	Most critical period. Organs form; heart beats; hands, fingers, feet, toes develop. Most vulnerable to damage.
2nd Trimester (15–28 wks)	Mother often most comfortable. Fetus can blink, hear conversations; regular sleep/wake cycles begin.
3rd Trimester (29–40 wks)	Rapid growth. Fetus uses all five senses. At week 40, weighs 6–9 lbs and is ready to be born.

What to Avoid During Pregnancy

- **Tobacco** — Linked to 30% of low-birth-weight babies, 14% of premature births. Vaping still exposes fetus to nicotine.
- **Alcohol** — NO safe amount during pregnancy. Causes Fetal Alcohol Spectrum Disorders (FASD) — leading preventable cause of intellectual disability in the U.S.
- **Drugs** — All medications need doctor approval. Illegal drugs cause severe risks: premature birth, developmental delays, addiction at birth.

Lesson 2 — Heredity and Genetics

Humans have **46 chromosomes in 23 pairs** in almost every cell. Each chromosome contains thousands of **genes** — the specific segments of DNA coding for individual traits. You share about 99.9% of your DNA with every other human on Earth; that 0.1% difference accounts for all the variation between people.

Gene Therapy — From Experimental to Real

When the original textbook was written, gene therapy was largely experimental. As of 2024, the FDA approved **eight new gene and cell therapies** in a single year, and over **76** have received global approval. **CRISPR-Cas9** is now the dominant gene editing tool. In 2023, the FDA approved the first CRISPR-based therapy (Casgevy) to treat sickle-cell disease.

Lesson 3 — Birth Through Childhood

Developmental psychologists divide life into stages, each with developmental tasks — milestones that need to happen for healthy growth.

Infancy (0–12 mo)	Fastest growth period. Brain forms 1 million+ new neural connections per second. Building trust is foundational.
Early Childhood (1–3 yrs)	Developing autonomy — confidence to control their own body and choices. 'No' becomes their favorite word.
Middle Childhood (4–6 yrs)	Initiating play, forming friendships, testing limits. Beginning to understand cause and effect.
Late Childhood (7–12 yrs)	School becomes central. Developing moral conscience and deeper friendships.

Chapter Vocabulary

Lesson 1

Fertilization	The union of a sperm and egg cell, resulting in a zygote; also called conception
Implantation	The process by which the zygote attaches to and burrows into the uterine wall
Embryo	The developing organism from weeks 3–8 of pregnancy
Fetus	The developing organism from week 9 through birth
Prenatal care	The health practices and medical supervision a pregnant woman undertakes to protect her health and her baby's
Fetal Alcohol Spectrum Disorders (FASD)	A range of permanent physical and cognitive conditions caused by prenatal alcohol exposure

Lesson 2

Chromosomes	Thread-like structures in the cell nucleus that carry the genetic code; humans have 46 (23 pairs)
Genes	Segments of chromosomes that are the basic units of heredity; each codes for specific traits
DNA	The double-helix molecule that makes up chromosomes and contains the complete genetic code
Genetic disorders	Conditions caused partly or completely by a defect or mutation in genes
Gene therapy	A medical treatment that corrects genetic disorders by introducing, repairing, or replacing defective genes

Lesson 3

Developmental tasks	The milestones and experiences that must occur at each life stage for healthy growth
Autonomy	The growing confidence and ability to control one's own body and choices